OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

# Semester II Examinations 2025–2026

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| **Course Instance Codes(s)** | 3BA1, 4BME1, 1EM1, 3BMS2, 4BMS2, 3BS9 |
| **Exam(s)** | 3rd Arts, 3rd Science, 4th Science, 4th BA Education |
| **Module Codes** | MA3491 |
| **Module** | **Fields and Applications** |
| Paper No | 1 |
| External Examiner | Dr M. Mackey |
| Internal Examiner(s) | Dr J. Cruickshank<br>⋆ Dr E. Sköldberg |

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| **Instructions:** | **Answer all FIVE questions** |
| **Duration** | 2 Hours |
| **Number of Pages** | 3 (including this page) |
| **Discipline** | Mathematics |

**Requirements:**

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| Release in Exam Venue | Yes ☑ No ☐ |
| Release to Library | Yes ☑ No ☐ |
| Statistical/ Log Tables: | State Examination Commission's *Formulae and Tables* |
| Handout | None |
| Cambridge Tables | None |
| Graph Paper | None |
| Log Graph Paper | None |
| Other Materials | None |
| Graphic material in colour | Yes ☐ No ☑ |

Q1. (a) (i) State the *sphere packing bound.* **[2 marks]**

(ii) Determine the upper bound that the Sphere Packing bound gives on the number of codewords in a binary code of block length 12 and minimum distance 5. **[4 marks]**

(b) (i) State the *Singleton bound.* **[2 marks]**

(ii) Prove the Singleton bound. **[4 marks]**

(iii) Determine the upper bound that the Singleton bound gives on the number of codewords in a binary code of block length 12 and minimum distance 5. **[2 marks]**

(c) Consider the block code over the alphabet $\{X, Y\}$ consisting of all words of length 8 where the number of $X$s that appear in the positions $1, 2, 3, 4$ is equal to the number of $Y$s that appear in positions $5, 6, 7, 8$. Determine the parameters of the code. **[6 marks]**

Q2. (a) Find all irreducible polynomials of degree 2 in $\mathbb{F}_2[x]$. **[3 marks]**

(b) Show that the polynomial $p(x) = x^5 + x^2 + 1 \in \mathbb{F}_2[x]$ is irreducible. **[5 marks]**

(c) Calculate the multiplicative inverse of $x^2 + 1$ in $\mathbb{F}_{32}$. (Use the presentation $\mathbb{F}_{32} = \mathbb{F}_2/(p(x))$ with $p(x)$ from part (b); give the result as a polynomial of degree $\leq 4$.) **[6 marks]**

(d) Let $F$ be a field of characteristic $p$. Show that, for all $a, b \in F$, we have

$$(a + b)^p = a^p + b^p.$$

**[6 marks]**

Q3. Consider the ternary code $C$ with generator matrix

$$G = \begin{pmatrix} 1 & 1 & 2 & 0 \\ 2 & 0 & 2 & 2 \end{pmatrix}.$$

(a) Construct a generator matrix in standard form for $C$. **[4 marks]**

(b) Construct a check matrix for $C$. **[2 marks]**

(c) Determine the minimum distance of $C$. **[2 marks]**

(d) Show that $C$ is self-dual, i.e. that $C = C^{\perp}$. **[4 marks]**

(e) (i) Construct a syndrome table for $C$. **[4 marks]**

(ii) Use the syndrome table to decode the message words `1111` and `1212`. **[4 marks]**

Q4. (a) What is the definition of the *Hamming code* $Ham(r, q)$? **[2 marks]**

(b) Construct a parity check matrix for the Hamming code $Ham(2, 4)$. **[4 marks]**

(c) Explain how a single error can be corrected in $Ham(2, 4)$. Decode the words `11111`, and `10101`. **[6 marks]**

(d) State (without proof) the parameters of the Hamming code $Ham(r, q)$. **[2 marks]**

(e) Show that the Hamming code $Ham(r, q)$ is a perfect code. **[6 marks]**

Q5. Let $C$ be the binary cyclic code of length 6 with generator polynomial $x^2 + x + 1$.

(a) Construct a generator matrix for $C$. **[4 marks]**

(b) Compute the check polynomial for $C$, and hence construct a check matrix for $C$. **[4 marks]**

(c) Define the *weight enumerator* of a linear code. **[2 marks]**

(d) Compute the weight enumerator for the code $C^\perp$. **[4 marks]**

(e) State the *MacWilliams identity*, and use it to compute the two terms of lowest degree of the weight enumerator of $C$. **[4 marks]**

(f) Hence, or otherwise, determine the minimum distance of $C$. **[2 marks]**